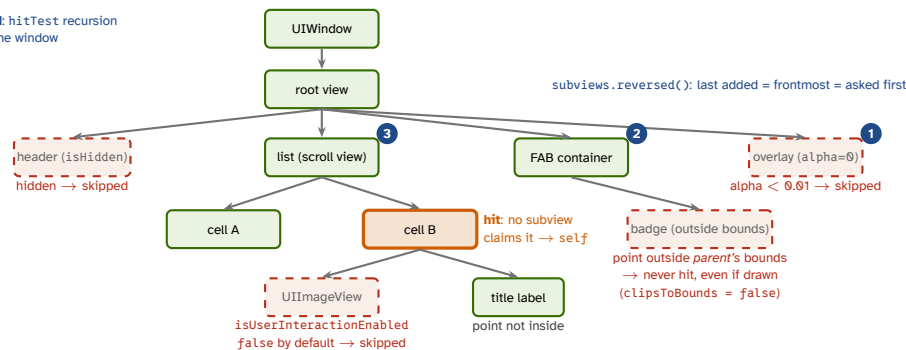
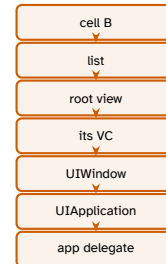


**In one line:** A touch goes **down** then **up**. On touch-down UIKit **hit-tests**: `hitTest(_:with:)` recurses from the window, **frontmost subview first**, and returns the **deepest** view whose `point(inside:with:)` is true — that view owns the touch for its whole life. **Gesture recognizers** on that view and its ancestors see the touch *before* the view does. Whatever is not handled — touches, key presses, nil-target actions — then walks **up** the responder chain via `next`.

DOWN: hitTest recursion from the window



then UP: next



## How it works

- Delivery:** `UIApplication.sendEvent` → `UIWindow.sendEvent`. Hit-test once at touch-down; `UITouch.view` stays the hit view even if the finger leaves it. The window gives the touch to **gesture recognizers first**, then to the view.
- Skipped by hit-testing:** `isHidden`, `isUserInteractionEnabled` = false (default on `UILabel`, `UIImageView` — and it disables the whole subtree), `alpha < 0.01`, and any point outside the view's bounds — a subview drawn outside its parent is visible but **untappable**. Transparent pixels still hit: content is not considered.
- Bigger tap target** (HIG: 44 × 44 pt): override `point(inside:with:)` with an outset rect — but it only works *inside every ancestor's bounds*.
- Touch methods:** `touchesBegan/Moved/Ended/Cancelled`. The default implementation **forwards up the chain**; override without super and you must override all four. Always handle *Cancelled* (a recognizer or a call can take the touch).
- First responder** receives keys, presses, edit-menu actions, nil-target actions. `becomeFirstResponder()` needs `canBecomeFirstResponder` and a window; a text input as first responder brings up the keyboard (inputView, inputAccessoryView replace/extend it); dismiss with `view.endEditing(true)`. `canPerformAction(_:withSender:)` gates edit-menu items; `UIKeyCommand` for hardware keys.
- Target-action:** `UIControl` does **not retain** its target; a nil target searches the responder chain. `sendAction(_:to:from:for:)` with `to: nil` starts at the first responder. Closure-based `addAction(UIAction)` (iOS 14) is retained — capture [weak self]. Custom controls: `begin/continue/endTracking(_:with:)`, `sendActions(for:)`.

## Gestures — the rules

- States:** discrete *possible* → *recognized* | *failed*; continuous *possible* → *began* → *changed*\* → *ended* | *cancelled*.
- Defaults:** `cancelsTouchesInView` = true (on recognition the view gets `touchesCancelled`), `delaysTouchesBegan` = false (view sees began in parallel), `delaysTouchesEnded` = true.
- Coordination:** by default two recognizers do not recognize together. `require(toFail:)` — single tap waits for double tap to fail (adds latency). Delegate: `gestureRecognizerShouldBegin`, `shouldRecognizeSimultaneouslyWith`, `shouldReceive(_ touch:)` (ignore touches on controls), `shouldRequireFailureOf` / `shouldBeRequiredToFailBy`.
- Controls vs an ancestor's recognizer:** the control's default action (button tap, slider drag) wins; a view can veto with `gestureRecognizerShouldBegin(_:)` (`UISlider` does). **Scroll view:** `delaysContentTouches` (true); `touchesShouldCancel(in:)` returns true unless the view is a `UIControl` — override to let a drag that starts on a button still scroll.

## Remember

**Down by hitTest, recognizers first, up by next.** Frontmost first · hidden, disabled, `alpha < 0.01`, out of bounds → skipped.

## Example — hit-testing, written out

```

// what the default does (a sketch of the documented rules)
override func hitTest(_ p: CGPoint, with e: UIEvent?) -> UIView? {
    guard !isHidden, isUserInteractionEnabled, alpha >= 0.01,
        point(inside: p, with: e) else { return nil }
    for sub in subviews.reversed() { // frontmost first
        if let hit = sub.hitTest(sub.convert(p, from: self), with: e) {
            return hit }
        return self }
    // 1 bigger tap area (still clipped by every ancestor's bounds)
    override func point(inside p: CGPoint, with e: UIEvent?) -> Bool {
        bounds.insetBy(dx: -12, dy: -12).contains(p) }
    // 2 pass-through overlay: its subviews get taps, empty areas don't
    override func hitTest(_ p: CGPoint, with e: UIEvent?) -> UIView? {
        let v = super.hitTest(p, with: e); return v == self ? nil : v }
}

```

## Custom view-controller transitions

- Modal:** `vc.transitioningDelegate` (**weak** — keep it alive) vends the animator (`animationController(forPresented:...)`); .custom style + a `UIPresentationController` for dimming and frame. **Push/pop:** the `UINavigationControllerDelegate` `animationControllerFor:from:to:method`.
- Animator** = `UIViewControllerAnimatedTransitioning`: `transitionDuration(using:)` + `animateTransition(using:)` — add the to view to `ctx.containerView`, `animate`, then **always** `ctx.completeTransition(!ctx.transitionWasCancelled)`; forget it and the UI stays frozen. `interruptibleAnimator(using:)` returns a `UIViewPropertyAnimator`.
- Interactive:** return a `UIPercentDrivenInteractiveTransition` from `interactionControllerFor...` *only while a gesture drives it*; the pan calls `update(%)`, then `finish()` or `cancel()` by progress + velocity.
- iOS 18:** `preferredTransition` = `.zoom { _ in thumb }` — system zoom, interactive throughout.

## Traps

- ! Tap gesture on `UIImageView/UILabel` never fires — interaction is off by default.
- ! Badge / button sticking out of its superview: drawn, never tappable.
- ! “Dismiss keyboard” tap recognizer on the root view kills `didSelectRowAt` — set `cancelsTouchesInView` = false.
- ! A clear full-screen view with `alpha 1` on top swallows every touch.
- ! `transitioningDelegate` is weak; no `completeTransition` → frozen UI.